WORKSHEET 4.3

Rates of Change Beyond Motion

Unit 4: Derivatives in Context

Key ideas

- Any derivative is a rate: of area with respect to side length, of population with respect to time, of cost with respect to quantity.
- **Marginal** cost, revenue or profit means the derivative with respect to the number of items.
- Read the independent variable carefully. $\dfrac{dA}{dr}$ is a rate per unit of length, not per unit of time.
- A quantity stops increasing or decreasing where its derivative changes sign.

Worked examples

Example 1 · Geometry

A square has side s cm and area $A = s^2$. Find the rate of change of the area with respect to the side when $s = 6$.

Solution. $\dfrac{dA}{ds} = 2s$.

12 cm^2 per cm.

Example 2 · Growth

A town has population $P(t) = 1200e^{0.05t}$ after t years. How fast is it growing at $t = 0$?

Solution. $P'(t) = 60e^{0.05t}$.

60 people per year.

Example 3 · Marginal cost

The cost of making x items is $C(x) = 500 + 8x + 0.02x^2$ dollars. Find the marginal cost when $x = 150$.

Solution. $C'(x) = 8 + 0.04x$.

14 dollars per item.

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Practice

CORE

1. A circle has circumference $C = 2\pi r$. Find $\dfrac{dC}{dr}$.

2. A cube has edge x and volume $V = x^3$. Find $\dfrac{dV}{dx}$ when $x = 2$.

3. A quantity is $Q(t) = 100 - 4t^2$. Find $Q'(3)$.

4. Pressure P is measured in kilopascals and volume V in litres. State the units of $\dfrac{dP}{dV}$.

EXAM LEVEL

5. A population of insects is $N(t) = \dfrac{600}{1 + 2t}$ after t weeks. Find $N'(2)$.

6. The temperature of a metal plate is $T(t) = 18 + 52e^{-0.05t}$ degrees Celsius after t minutes. How fast is it cooling at $t = 0$?

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7. The profit from selling x items is $P(x) = 30x - 0.05x^2 - 200$ dollars. At what value of x is the marginal profit zero?

8. A cylinder has fixed height 10 cm and radius r cm. Find the rate of change of its volume with respect to its radius when $r = 3$.

Question 9 · Free-response practice

The mass of a bacterial culture is $M(t) = 40 + 10t - t^2$ grams for $0 \le t \le 8$ hours.

(a) Find $M'(2)$. Indicate units of measure.

(b) At what time is the mass of the culture greatest, and what is that mass?

(c) Find the average rate of change of the mass over $0 \le t \le 8$, and the time at which the instantaneous rate of change equals it.

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Worked solutions

1. 2π.
2. $3x^2 = 12$.
3. $Q'(t) = -8t$, so $Q'(3) = -24$.
4. Kilopascals per litre.
5. $N'(t) = -\dfrac{1200}{(1+2t)^2}$, so $N'(2) = -48$ insects per week.
6. $T'(t) = -2.6e^{-0.05t}$, so it is cooling at 2.6 degrees Celsius per minute.
7. $P'(x) = 30 - 0.1x = 0$ at $x = 300$.
8. $V = 10\pi r^2$, so $\dfrac{dV}{dr} = 20\pi r = 60\pi$ cm^3 per cm.
9. (a) $M'(t) = 10 - 2t$, so $M'(2) = 6$ grams per hour.
   (b) $M'(t) = 0$ at $t = 5$, where M' changes from positive to negative. $M(5) = 65$ grams.
   (c) $\dfrac{M(8) - M(0)}{8} = \dfrac{56 - 40}{8} = 2$ grams per hour. $10 - 2t = 2$ at $t = 4$.

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